

Question:
Coical representation of body in the cerebrum is
Answer Choices:
A. Horizontal
B. Veical
C. Tandem
D. Oblique

Reasoning + Answer:

<tool_call>

The cerebral representation of the body in the cere brum , specifically in the somatosensory cortex , follows a topographic map known as the somatotopic organization or "homunculus." This map is primarily organized in a vertical (or vertical-like) arrangement , where different body parts are represented in a vertical column . The face is at the top , the trunk in the middle , and the legs and arms are at the bottom . The arrangement is not horizontal , tandem , or oblique in the sense of a simple directional orientation ; rather , it is vertically oriented with respect to the cortical surface .

In the primary somatosensory cortex of the postcentral gyrus , the body is represented in a vertical layout . This is known as the "cortical homunculus," where the size of the representation correlates with sensory sensitivity âĖe.g ., the face and hands are relatively large due to high sensory acuity .

The term "vertical" here refers to the orientation of the body representation in the cortical plane , not in a strict anatomical vertical axis . The representation is vertically organized from top to bottom (face to feet), which makes "vertical" the most accurate choice among the options .

- Option A (Horizontal) is incorrect because the body is not laid out side to side .
- Option C (Tandem) is not a standard term for cortical body representation .
- Option D (Oblique) implies a diagonal pattern , which is not the case .
- Option B (Vertical) correctly describes the orientation of the body map .

Therefore , the correct answer is vertical , as the body is represented vertically in the somatosensory cortex .

< answer >

B . Veical

</ answer >

